

✓ Power Function

```
%%writefile function1.cpp
```

```
#include<iostream>
using namespace std;

int power(){
    int a,b;
    cin>>a>>b;
    int ans=1;
    for(int i=1;i<=b;i++){
        ans=ans*a;
    }
    return ans;
}

int main(){
    // int a, b;
    // cin>>a>>b;
    // int answer=power(a,b);
    int ans=power();
    cout<<"answer is : "<<ans<<endl;
    return 0;
}
```

 Writing function1.cpp

```
!g++ function1.cpp -o function1
!./function1
```

 13 2
answer is : 169

✓ Given Number is even or odd

```
%%writefile function2.cpp
```

```
#include<iostream>
using namespace std;

int evenodd(int a){
    if (a%2==0){
        return 1;
    }
    else{
        return 0;
    }
}
```

```
int main(){
    int a;
    cin>>a;
    if (evenodd(a)){
        cout<<"even"<<endl;
    }
    else{
        cout<<"odd"<<endl;
    }
    return 0;
}
```

 Writing function2.cpp

```
!g++ function2.cpp -o function2
!./function2
```

 23
odd

✓ Finding Combination

```
%%writefile function3.cpp

#include<iostream>
using namespace std;

int fact(int n){
    int fact=1;
    for (int i=1;i<=n;i++){
        fact*=i;
    }
    return fact;
}

int nCr(int n,int r){
    int num=fact(n);
    int denom=fact(r)*fact(n-r);
    int ans=num/denom;
    return ans;
}

int main() {
    int n,r;
    cin >> n>>r;
    cout<<"answer is : "<<nCr(n,r)<<endl;;
}


```

 Writing function3.cpp

```
!g++ function3.cpp -o function3
!./function3
```

 12 3
answer is : 220

✓ Finding Permutation

```
%%writefile function4.cpp

#include<iostream>
using namespace std;

int fact(int n){
    int fact=1;
    for (int i=1;i<=n;i++){
        fact*=i;
    }
    return fact;
}

int nPr(int n,int r){
    return fact(n)/fact(n-r);
}

int main() {
    int n,r;
    cin >> n>>r;
    cout<<"answer is : "<<nPr(n,r)<<endl;;
}


```

 Writing function4.cpp

```
!g++ function4.cpp -o function4
!./function4
```

 12 3
answer is : 1320

✓ Print Counting

```
%%writefile function5.cpp

#include<iostream>
using namespace std;

//function signature
```

```

void printCounting(int n){
    //function body
    for(int i=1;i<=n;i++){
        cout<<i<<" ";
    }
    cout<<endl;
}
int main(){
    int n;
    cin>>n;
    //function body
    printCounting(n);
}

```

 Writing function5.cpp

```

!g++ function5.cpp -o function5
!./function5

```

 12
1 2 3 4 5 6 7 8 9 10 11 12

✓ Given number is prime or not

```
%%writefile function6.cpp
```

```

#include<iostream>
using namespace std;

//1-> prime number
//0-> not prime

bool isPrime(int n){
    for(int i=2;i<n;i++){
        if (n%i==0){
            return 0;
        }
    }
    return 1;
}
int main(){
    int n;
    cin>>n;
    cout<<isPrime(n);
}

```

 Writing function6.cpp

```

!g++ function6.cpp -o function6
!./function6

```

 13
1

✓ Find the nth term of an AP

```
%%writefile function7.cpp
```

```

#include<iostream>
using namespace std;

int nthTerm(int n){
    int ans = 3*n+7;
    return ans;
}
int main(){
    int n;
    cin>>n;
    cout<<"nth term is "<<nthTerm(n);
}

```

 Writing function7.cpp

```

!g++ function7.cpp -o function7
!./function7

```

 2
nth term is 13

✓ Fibonacci Seies

```
%%writefile function8.cpp

#include<iostream>
using namespace std;

void fiboSeries(int n){
    int sum =0 , t1=0 , t2=1;
    for(int i=0;i<=n;i++){
        sum=t1+t2;
        t1=t2;
        t2=sum;
        cout<<sum<<" ";
    }
    cout<<endl;
}

int main(){
    int n;
    cin>>n;
    //function body
    fiboSeries(n);
}
```

 Writing function8.cpp

```
!g++ function8.cpp -o function8
!./function8
```

 6
1 2 3 5 8 13 21

✓ Set Bit

```
%%writefile function9.cpp

#include<iostream>
using namespace std;

int main(){
    int n;
    cin>>n;
    int count=0;
    while(n!=0){
        if (n&1){
            count+=1;
        }
        n=n>>1;
    }
    cout<<count;
}
```

 Overwriting function9.cpp

```
!g++ function9.cpp -o function9
!./function9
```

 12
2